RFI DETECTION IN SENTINEL-1 SAR IMAGERY VIA PSEUDO-LABEL DISTILLATION AND MULTI-RESOLUTION ENSEMBLE

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ABSTRACT

We describe the *UdeM AI* team's solution to the ESA ClearSAR Track 1 Grand Challenge: Radio-Frequency Interference (RFI) detection in Sentinel-1 SAR quicklooks. Our pipeline combines a YOLO26x base detector with pixel-space Normalised Wasserstein Distance (NWD) regression, pseudo-label distillation from a multi-source ensemble teacher onto unlabelled test imagery, multi-resolution test-time augmentation (TTA) fused by Weighted Box Fusion (WBF), and a selective cross-fold ensemble that drops the domain-overlapping fold. Ablation on a 401-image leak-isolated meta-validation set shows pseudo-distillation contributes $+0.041$ mAP$_{50\text{-}95}$ over the per-fold baseline, multi-resolution TTA adds $+0.034$, and selective fold dropping adds $+0.002$, for a total $+0.077$ over the per-fold meta-validation baseline. The final submission reaches 0.4776 mAP$_{50\text{-}95}$ on the held-out test set, placing 28th of 172 teams. All training and fine-tuning used a single cloud NVIDIA A100 GPU.

Index Terms— Ensemble, pseudo-label distillation, radio-frequency interference, SAR imagery, small object detection.

1. TASK AND DATA

The ESA ClearSAR Track 1 Grand Challenge [1] frames RFI detection in Sentinel-1 quicklooks as a COCO-style object detection task with mAP averaged over IoU 0.50–0.95. The labelled set contains 3,154 images; 786 test images are held out by the organisers and scored by the challenge platform from a zipped COCO-format JSON of detections. RFI appears as thin horizontal stripes (median height $\approx$10 px, median width $\approx$140 px) in variable-size RGB quicklooks (median $\approx$515$\times$342 px), with a median per-box aspect ratio of $\approx$8:1. Three constraints follow: IoU-based regression gives no gradient for non-overlapping thin boxes; SAR side-looking geometry breaks horizontal symmetry, so flip TTA is disabled (H-flip costs -6.5% [2]); and inference resolutions above 640 px amplify speckle relative to signal. We reserve a leak-isolated meta-validation set of 401 images sampled from the fold-0 validation pool (seed 42) for honest local evaluation.

2. METHOD

The pipeline (Figure 1) has four stages.

Base detector. YOLO26x [3] ($\approx$55.7 M parameters, COCO pre-trained, no P2 head) trained with a CIoU + NWD box loss $\mathcal{L}_{\text{box}} = (1-\alpha)\mathcal{L}_{\text{CIoU}} + \alpha\mathcal{L}_{\text{NWD}}$ at α=0.5, with NWD [4] computed in pixel space (normalisation constant C=12.8 px). Ultralytics' `BboxLoss.forward` receives boxes in stride-relative grid units, so we multiply by the anchor stride before computing NWD; this keeps gradients non-saturating for the small heights characteristic of RFI ($\approx$10 px median). Images are pre-processed with CLAHE (clip 3.0, 8×8 tiles) on the LAB L-channel. Horizontal and vertical flips are disabled. Training uses MuSGD for up to 120 epochs (patience 20) at 640 px, batch 16, η_0=10^{-3} with cosine decay to 10^{-5}.

Pseudo-label distillation. The teacher (V12) is our strongest pre-distillation ensemble: a single-stage WBF [5] over multiple YOLO26x fold and multi-resolution-TTA sources together with one RF-DETR-Large [6] fold checkpoint at weight 1.5$\times$, fused with τ=0.70 and `conf_type`=max. V12 scores 0.4755 on the held-out test set. The teacher is applied to the 786 unlabelled test images with confidence threshold 0.75 (max 20 boxes/img), producing 600 pseudo-labelled images. For each fold $f \in \{1, 2, 4\}$, a student is fine-tuned from the per-fold base on the fold-augmented dataset with AdamW [7] (η_0=10^{-4}, `freeze`=10, 15 epochs, batch 24, same NWD blend). Fold 0 is omitted because its base partition overlaps the meta-validation pool.

Multi-resolution TTA and fusion. Each student is inferred at $\mathcal{I} = \{608, 640, 672, 704\}$ px; WBF across resolutions per fold yields one TTA source. The three per-fold TTA sources are then fused with equal weights into a super-source S^*. The final submission V17 is the two-source WBF blend $D_{\text{final}} = \text{WBF}(D_{\text{V12}}, S^*; w{=}(1.0, 0.25), \tau{=}0.70)$.

Table 1. Component ablation on the 401-image meta-validation set.

Configuration	**mAP**$_{50\text{-}95}$	Δ
Per-fold baseline (fold 0)	0.394	—
+ Pseudo-FT, fold 4 only	0.435	$+0.041$
+ Multi-res TTA, fold 4 only	0.469	$+0.034$
+ 3-fold (drop f_0), equal weights	0.471	$+0.002$
+ V12 blend @ w=0.25 (V17)	0.471†	—

†Meta-val cannot exceed V12 by adding an honest source (V12 is leaked on this split). Official held-out test: **0.4776**.

3. RESULTS AND DISCUSSION

Table 1 reports the component ablation. Pseudo-distillation contributed the largest single gain ($+0.041$): the soft signal in the teacher's confidence scores compensates for the small training set more effectively than the architectural alternatives we evaluated. Multi-resolution TTA added a further $+0.034$ because thin stripes a few pixels tall are resolved unevenly by YOLO's strided feature pyramid at any single input size. Dropping the domain-overlapping fold-0 from the cross-fold ensemble was net-positive: correlated predictions drag the WBF consensus when one source has seen the validation data during training. Blending the V12 teacher into the final fusion does not improve the leaked meta-val baseline but transfers cleanly

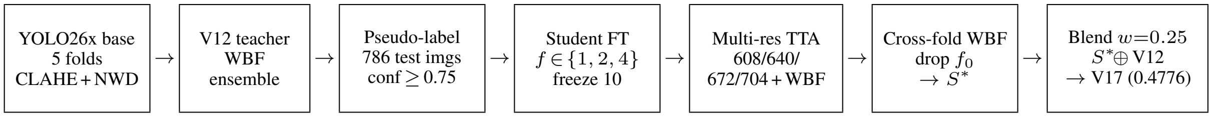

Fig. 1. Pipeline. A YOLO26x base detector (CLAHE input, pixel-space NWD loss) is trained per fold; the V12 pre-distillation ensemble pseudo-labels the 786 test images; per-fold students are fine-tuned on the fold-augmented data, run multi-resolution TTA fused by WBF, fused across folds (dropping the domain-overlapping fold 0) into a super-source S^*, and blended with V12 at w=0.25.

to held-out test (V12 alone = $0.4755 \rightarrow$ V17 = 0.4776).

Negative results. A second architectural family (RF-DETR) plateaued at $+0.0002$ across six checkpoint–weight variants; D-FINE-X [8] fine-tuned on fold 0 for 10 epochs reached 3×10^{-4} mAP, suggesting DETR-family decoders with different matching [9] would need substantially more epochs to adapt to the elongated geometry of RFI; inference-only SAHI [10] at 320-px tiles regressed -0.085, fragmenting long horizontal stripes at tile boundaries; and temperature-scaled calibration was at most $+0.0002$.

4. CONCLUSION

The final pipeline reaches 0.4776 mAP$_{50\text{-}95}$ on the held-out test set; the meta-validation ablation chain (Table 1) totals $+0.077$ over the per-fold baseline. The submission placed the *UdeM AI* team 28th of 172 teams on the public leaderboard of ESA ClearSAR Track 1. All training and fine-tuning were performed on a single cloud NVIDIA A100 GPU. The largest remaining gap is at high IoU thresholds, where sub-pixel height accuracy dominates and a 640-px input under-resolves the median ≈10-px stripe; sliced *training* at 320-px tiles is the most promising next move.

5. REFERENCES

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